CLINICAL STUDY REPORT

A Phase 3 Randomised, Placebo-controlled Study of Kelvaparib in Recurrent High-grade Serous Ovarian Carcinoma

Study KLV-7703

Investigational product	Kelvaparib (KLV-3308), oral PARP1-selective inhibitor
Indication studied	High-grade serous ovarian carcinoma
Development phase	Phase 3
Sponsor	Marnholt Oncology GmbH
Protocol number	KLV-7703
First subject enrolled	28 March 2020
Data cut-off date	10 January 2023
Database lock	7 February 2023
Report date	24 July 2023
Report status	Final
Good Clinical Practice	Conducted in accordance with the Declaration of Helsinki and ICH Good Clinical Practice.

2 SYNOPSIS

Name of sponsor	Marnholt Oncology GmbH
Name of finished product	KLV-3308
Name of active ingredient	Kelvaparib (KLV-3308)
Title of study	A Phase 3 Randomised, Placebo-controlled Study of Kelvaparib in Recurrent High-grade Serous Ovarian Carcinoma
Study number	KLV-7703
Phase of development	Phase 3
Studied period	28 March 2020 to 10 January 2023
Analysis	Final analysis
Objectives	Primary: progression-free survival by investigator assessment. Secondary: duration of response, overall survival and the incidence and severity of treatment-emergent adverse events, and pharmacokinetics.
Methodology	An open-label, multicentre, randomised study allocating subjects 2:1 to Kelvaparib 300 mg or Placebo. Randomisation is stratified by BRCA mutation status, platinum sensitivity and residual disease after surgery. Approximately 354 subjects.
Number of subjects	354 subjects were randomised and received at least one dose of study treatment (236 Kelvaparib 300 mg, 118 Placebo).
Diagnosis and main criteria for inclusion	Adults with histologically or cytologically confirmed recurrent ovarian carcinoma, platinum-sensitive or platinum-resistant, measurable disease assessable by RECIST version 1.1, ECOG performance status 0 to 2, adequate organ function.
Test product, dose and mode of administration	Kelvaparib, film-coated tablets, administered orally twice daily on a continuous schedule, with weekly blood counts for the first month and monthly thereafter.
Duration of treatment	Until disease progression, unacceptable toxicity or withdrawal of consent. Median duration 132 days.
Criteria for evaluation - efficacy	Best overall response by RECIST version 1.1, and duration of response, progression-free survival, overall survival.
Criteria for evaluation - safety	Treatment-emergent adverse events graded by CTCAE version 5.0, laboratory parameters and vital signs.
Statistical methods	The primary endpoint is compared between treatment groups using a stratified log-rank test at a two-sided significance level of 0.05, with the hazard ratio and its 95 per cent confidence interval from a stratified Cox model. Key secondary endpoints are tested in a prespecified hierarchical order; significance is claimed for a secondary endpoint only if every preceding test in the hierarchy reached significance. Approximately 219 events provide 90 per cent power to detect a hazard ratio of 0.63.
Summary - efficacy	The overall response rate was Kelvaparib 300 mg 129 of 227 subjects (56.8 per cent, 95 per cent CI (50.1, 63.4)); Placebo 28 of 109 subjects (25.7 per cent, 95 per cent CI (17.8, 34.9)). Complete response was reported in 30 subjects (8.9 per cent) overall. Median progression-free survival 8.8 months.

Summary - safety	Grade 3 or higher events in 179 subjects (50.6 per cent), serious events in 83 (23.4 per cent). Nausea was the most frequent event (223 subjects, 63.0 per cent). Myelodysplastic syndrome or acute myeloid leukaemia, an event of special interest, occurred in 8 subjects (2.3 per cent). 43 discontinued because of an adverse event; 72 deaths were reported.
Conclusion	Kelvaparib did not prolong progression-free survival compared with placebo in recurrent ovarian carcinoma, platinum-sensitive or platinum-resistant: progression-free survival was shorter with kelvaparib (stratified hazard ratio 1.77, 95 per cent CI 1.26 to 2.50; stratified log-rank p = 0.0009), and the primary endpoint was not met. The overall response rate of 46.7 per cent and every other secondary result are descriptive, because the testing hierarchy stops at the primary endpoint.
Date of report	24 July 2023

3 TABLE OF CONTENTS

Section	Title	Page
1	Title page	1
2	Synopsis	2
3	Table of contents	4
4	List of abbreviations and definitions of terms	6
5	Ethics	6
5.1	Independent ethics committee or institutional review board	6
5.2	Ethical conduct of the study	7
5.3	Subject information and consent	7
6	Investigators and study administrative structure	7
7	Introduction	7
8	Study objectives	7
9	Investigational plan	8
9.1	Overall study design and plan	8
9.2	Discussion of study design, including choice of control	8
9.3	Selection of study population	8
9.4	Treatments	8
9.5	Efficacy and safety variables	8
9.6	Data quality assurance	9
9.7	Statistical methods planned in the protocol	9
9.8	Changes in the conduct of the study or the planned analyses	9
10	Study subjects	10
10.1	Disposition of subjects	10
10.1.1	Reasons for discontinuation	10
10.2	Protocol deviations	11
11	Efficacy evaluation	11
11.1	Data sets analysed	11
11.2	Demographic and other baseline characteristics	12
11.2.1	Discussion of each baseline characteristic	13
11.3	Measurements of treatment compliance	13
11.4	Efficacy results	15
11.4.1	Analysis of the primary endpoint	15
11.4.2	Analysis of time-to-event endpoints	16
11.4.3	Subgroup analyses	16
11.4.4	Efficacy conclusions	19
11.5	By-subject efficacy data	19
12	Safety evaluation	19
12.1	Extent of exposure	19

Section	Title	Page
12.2	Adverse events	19
12.2.1	Overview	19
12.2.1.1	Discussion of each event term	20
12.2.2	Events by system organ class and preferred term	26
12.2.3	Events of grade 3 or higher	29
12.2.4	Events considered related to study treatment	30
12.2.5	Timing of onset	31
12.2.6	Adverse events of special interest	31
12.3	Deaths, other serious adverse events and other significant events	33
12.3.1	Deaths	33
12.3.2	Serious adverse events	34
12.3.3	Events leading to discontinuation	35
12.3.4	Narratives	35
12.4	Clinical laboratory evaluation	35
12.4.1	Findings for each laboratory test	36
12.5	Vital signs, physical findings and other safety observations	38
12.5.1	Electrocardiograms	38
12.6	Safety conclusions	38
13	Discussion and overall conclusions	39
14	Tables and figures referred to but not in the text	41
14.1	Demographic data	41
14.2	Efficacy data	42
14.3	Safety data	51
14.3.10	Adverse events within subgroups	62
14.4	Pharmacokinetic data	69
15	Reference list	71
16	Appendices	72

4 LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation	Definition
ALT	alanine aminotransferase
AST	aspartate aminotransferase
AUC0-12	area under the concentration-time curve over one dosing interval (0 to 12 hours)
AUC0-24	area under the concentration-time curve from 0 to 24 hours
BRCA	breast cancer susceptibility gene
BRCA1	breast cancer susceptibility gene 1
BRCA2	breast cancer susceptibility gene 2
CA-125	cancer antigen 125
CI	confidence interval
CL/F	apparent oral clearance
Cmax	maximum plasma concentration
CR	complete response
CTCAE	Common Terminology Criteria for Adverse Events
Ctrough	trough plasma concentration
CV	coefficient of variation
ECOG	Eastern Cooperative Oncology Group
ICH	International Council for Harmonisation
KLV-3308	kelvaparib (compound code)
MedDRA	Medical Dictionary for Regulatory Activities
NE	not estimable
PFS	progression-free survival
PR	partial response
Q1	first quartile
Q3	third quartile
RECIST	Response Evaluation Criteria in Solid Tumours
SD	stable disease
t1/2	terminal elimination half-life
TEAE	treatment-emergent adverse event
tmax	time to maximum plasma concentration

5 ETHICS

5.1 Independent ethics committee or institutional review board

The protocol, its amendments, the subject information sheet and the consent form were reviewed and approved by the responsible committee for every participating site before any subject was enrolled at that site. Approvals are held in the trial master file.

5.2 Ethical conduct of the study

The study was conducted in accordance with the protocol, the principles of the Declaration of Helsinki, the principles of Good Clinical Practice and the applicable regulatory requirements.

5.3 Subject information and consent

Written informed consent was obtained from every subject before any study-specific procedure. Where the information sheet was revised during the study, consent was obtained again from subjects still participating.

6 INVESTIGATORS AND STUDY ADMINISTRATIVE STRUCTURE

The study was conducted at the sites listed in Appendix 16.1.4, identified by number. Investigator names are not recorded in this report. The sponsor was responsible for the design of the study, for the statistical analysis and for the preparation of this report. Clinical monitoring and data management were performed by Steinbeck Clinical Research GmbH. Laboratory analyses were performed at a central laboratory. Response and progression were assessed by the investigator; no independent review was performed.

7 INTRODUCTION

This report describes study KLV-7703, A Phase 3 Randomised, Placebo-controlled Study of Kelvaparib in Recurrent High-grade Serous Ovarian Carcinoma. The study was conducted to evaluate Kelvaparib in subjects with recurrent ovarian carcinoma, platinum-sensitive or platinum-resistant. The rationale for the study, for the dose and for the choice of endpoints is given in the protocol.

8 STUDY OBJECTIVES

Table 8-1 Study objectives

Type	Objective
Primary	Progression-free survival by investigator assessment
Safety	To characterise the safety and tolerability of Kelvaparib.

9 INVESTIGATIONAL PLAN

9.1 Overall study design and plan

An open-label, multicentre, randomised study allocating subjects 2:1 to Kelvaparib 300 mg or Placebo. Randomisation is stratified by BRCA mutation status, platinum sensitivity and residual disease after surgery. Approximately 354 subjects.

9.2 Discussion of study design, including choice of control

The design was appropriate to the question. A concurrent randomised control was used so that the treatment effect could be estimated without relying on an external comparison. The assessment interval was chosen so that the endpoint could be observed without imposing avoidable burden, and it was applied consistently across the study.

9.3 Selection of study population

Eligibility was defined in the protocol and is summarised in the synopsis (Section 2). Every subject was required to meet all inclusion criteria and none of the exclusion criteria; no waivers were granted.

9.4 Treatments

Kelvaparib was administered orally twice daily on a continuous schedule. Dose modification followed the rules in the protocol. Study treatment was supplied by the sponsor and accountability was reconciled at each monitoring visit.

9.5 Efficacy and safety variables

Efficacy was assessed according to RECIST version 1.1. Safety was assessed by adverse event reporting, laboratory testing, vital signs and physical examination, with severity graded using the Common Terminology Criteria for Adverse Events and events coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 25.0; concomitant medications were coded using the WHO Drug Dictionary (WHODrug) Global B3, March 2022. Medical history, physical examination, electrocardiogram, coagulation, urinalysis and pregnancy test results were recorded at the sites but are not held in the study database and are not reported.

9.6 Data quality assurance

Data were recorded on an electronic case report form with programmed edit checks. Source data verification covered consent, eligibility, the primary endpoint and all serious adverse events for every subject. Coding was performed centrally and reviewed by the sponsor's safety physician. The database was locked after all queries were resolved and the analysis populations were agreed.

9.7 Statistical methods planned in the protocol

The primary endpoint is compared between treatment groups using a stratified log-rank test at a two-sided significance level of 0.05, with the hazard ratio and its 95 per cent confidence interval from a stratified Cox model. Key secondary endpoints are tested in a prespecified hierarchical order; significance is claimed for a secondary endpoint only if every preceding test in the hierarchy reached significance. Approximately 219 events provide 90 per cent power to detect a hazard ratio of 0.63.

9.8 Changes in the conduct of the study or the planned analyses

The protocol was first approved as version 1.0 on 14 March 2020. It was amended 3 times before the data cut-off of 10 January 2023:

Amendment 1 (protocol version 2.0, 24 March 2021): Clarification of eligibility and monitoring. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. No change to the analysis populations or to the primary endpoint is introduced by this amendment.

Amendment 2 (protocol version 3.0, 9 October 2021): Revision following emerging safety data. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. No change to the analysis populations or to the primary endpoint is introduced by this amendment.

Amendment 3 (protocol version 4.0, 26 April 2022): Administrative and statistical update. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. The revised sample size is reflected in the statistical analysis plan, which is updated in parallel with this amendment.

The protocol of record for this report is version 4.0 (amendment 3, 26 April 2022).

Changes to the planned analyses were documented in the statistical analysis plan before database lock. No analysis reported here was devised after the data were examined.

10 STUDY SUBJECTS

10.1 Disposition of subjects

354 subjects were enrolled and treated. Disposition is in Table 10-1. Of these, 272 discontinued study treatment, most commonly for disease progression.

Table 10-1 Subject Disposition – Safety Analysis Set

Disposition	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Subjects treated, n	236	118	354
Treatment ongoing at data cut-off, n (%)	61 (25.8)	21 (17.8)	82 (23.2)
Discontinued study treatment, n (%)	175 (74.2)	97 (82.2)	272 (76.8)
Disease progression	40 (16.9)	26 (22.0)	66 (18.6)
Adverse event	11 (4.7)	6 (5.1)	17 (4.8)
Death	20 (8.5)	8 (6.8)	28 (7.9)
Withdrawal by subject	37 (15.7)	24 (20.3)	61 (17.2)
Physician decision	35 (14.8)	19 (16.1)	54 (15.3)
Lost to follow-up	17 (7.2)	6 (5.1)	23 (6.5)
Protocol deviation	15 (6.4)	8 (6.8)	23 (6.5)

Percentages are based on the number of subjects treated within each group.

10.1.1 Reasons for discontinuation

Disease progression. Recorded for 66 subjects (18.6 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Withdrawal by subject. Recorded for 61 subjects (17.2 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Physician decision. Recorded for 54 subjects (15.3 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Death. Recorded for 28 subjects (7.9 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Lost to follow-up. Recorded for 23 subjects (6.5 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Protocol deviation. Recorded for 23 subjects (6.5 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Adverse event. Recorded for 17 subjects (4.8 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

10.2 Protocol deviations

No protocol deviation dataset was collected in this study, and deviations were not classified as important or not important. The deviations held in the data are those recorded as the reason for discontinuing study treatment: 23 subjects (6.5 per cent of those treated, Table 10-1). These subjects are listed at Appendix 16.2.2. No monitoring record or deviation log is held in the study database, so deviations that did not lead to discontinuation of study treatment cannot be identified or listed.

11 EFFICACY EVALUATION

11.1 Data sets analysed

Table 11-1 Analysis sets

Analysis set	Subjects	Definition
Enrolled	354	All subjects assigned to treatment.
Safety	354	All subjects who received at least one dose.
Full analysis	336	All subjects in the safety analysis set with at least one post-baseline disease assessment.
Efficacy evaluable	336	Subjects with at least one evaluable post-baseline assessment.

11.2 Demographic and other baseline characteristics

Table 11-2 Demographic and Baseline Characteristics – Safety Analysis Set

Characteristic	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Age (years)			
Mean (SD)	65.0 (8.44)	66.4 (8.93)	65.5 (8.62)
Median	65.0	66.5	65.0
Min, max	44, 88	45, 88	44, 88
>= 65 years, n (%)	125 (53.0)	69 (58.5)	194 (54.8)
Sex, n (%)			
Male	0 (0.0)	0 (0.0)	0 (0.0)
Female	236 (100.0)	118 (100.0)	354 (100.0)
Race, n (%)			
White	192 (81.4)	98 (83.1)	290 (81.9)
Asian	19 (8.1)	6 (5.1)	25 (7.1)
Black or African American	16 (6.8)	6 (5.1)	22 (6.2)
Other	9 (3.8)	8 (6.8)	17 (4.8)
ECOG performance status, n (%)			
0	98 (41.5)	50 (42.4)	148 (41.8)
1	118 (50.0)	53 (44.9)	171 (48.3)
2	20 (8.5)	15 (12.7)	35 (9.9)
BRCA mutation status, n (%)			
BRCA1 mutated	55 (23.3)	29 (24.6)	84 (23.7)
BRCA2 mutated	44 (18.6)	14 (11.9)	58 (16.4)
BRCA wild-type	137 (58.1)	75 (63.6)	212 (59.9)
Homologous recombination deficiency, n (%)	122 (51.7)	63 (53.4)	185 (52.3)
Platinum-sensitive disease, n (%)	155 (65.7)	82 (69.5)	237 (66.9)
Residual disease after cytoreductive surgery, n (%)	101 (42.8)	46 (39.0)	147 (41.5)
Number of prior lines of platinum-based therapy			
Median	2.0	2.0	2.0
>= 3, n (%)	64 (27.1)	33 (28.0)	97 (27.4)

SD = standard deviation; ECOG = Eastern Cooperative Oncology Group. Percentages are based on the number of subjects in the safety analysis set within each group.

11.2.1 Discussion of each baseline characteristic

Age. The median age was 65 years, with a range of 44 to 88. 194 subjects (54.8 per cent) were aged 65 or over. The age distribution is consistent with the population the eligibility criteria describe and no age-related restriction was applied beyond the requirement to be an adult.

Sex. All 354 subjects were female, as the eligibility criteria require.

Race. White: 290 (81.9 per cent); Asian: 25 (7.1 per cent); Black or African American: 22 (6.2 per cent); Other: 17 (4.8 per cent). Race was self-reported and was recorded in the categories used across the programme. The distribution reflects the territories in which the study recruited and should be borne in mind when generalising these results.

Performance status at baseline. score 0: 148 (41.8 per cent); score 1: 171 (48.3 per cent); score 2: 35 (9.9 per cent). Performance status is the single strongest predictor of tolerance in this population, and the distribution here is what the eligibility criteria were intended to produce.

Prior therapy. The median number of prior lines was 2, with a range of 1 to 4. Subjects entering with more prior therapy have by definition disease that has already been selected by exposure to more mechanisms, and the analyses by this characteristic are presented in the subgroup section.

BRCA mutation status. BRCA wild-type: 212 (59.9 per cent); BRCA1 mutated: 84 (23.7 per cent); BRCA2 mutated: 58 (16.4 per cent). Recorded at baseline and prespecified as a subgroup.

Homologous recombination status. Present in 185 subjects (52.3 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

Platinum sensitivity. Present in 237 subjects (66.9 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

Residual disease after surgery. Present in 147 subjects (41.5 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

11.3 Measurements of treatment compliance

The median duration of exposure was 132 days and the median relative dose intensity was 85.8 per cent. Exposure is summarised in Table 11-3.

Table 11-3 Extent of Exposure – Safety Analysis Set

Exposure parameter	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Duration of treatment (days)			
Median	134.0	127.5	131.5
Q1, Q3	82.0, 181.8	73.8, 165.5	80.0, 179.0
Min, max	8, 307	3, 280	3, 307
Cycles started, median	5.0	4.5	5.0
Relative dose intensity (%), median	85.7	85.9	85.8
Any dose reduction, n (%)	42 (17.8)	23 (19.5)	65 (18.4)
Any dose interruption, n (%)	51 (21.6)	32 (27.1)	83 (23.4)

Q1 = first quartile; Q3 = third quartile. Relative dose intensity is the total dose received divided by the total dose planned over the treatment period.

All 354 subjects who were enrolled received at least one dose and are included in the safety population. The median duration of exposure was 132 days, with a range of 3 to 307.

The median relative dose intensity was 85.8 per cent. This is the measure that distinguishes a regimen that is genuinely tolerable from one that appears tolerable only because subjects are missing doses, and a value at this level indicates that most subjects received most of what was planned.

65 subjects (18.4 per cent) had at least one dose reduction and 83 (23.4 per cent) had at least one interruption. Reductions and interruptions followed the rules in the protocol; no subject was re-escalated after a reduction.

Exposure was sufficient to characterise both the efficacy and the safety of the regimen. A study in which most subjects stop early cannot support conclusions about either, and that is not the case here.

11.4 Efficacy results

11.4.1 Analysis of the primary endpoint

The primary endpoint was progression-free survival, defined as the time from the first dose of study treatment to the first documented disease progression by investigator assessment or death from any cause, whichever occurs first. 148 of 227 subjects in the Kelvaparib 300 mg group and 55 of 109 in the Placebo group had a progression-free survival event; median progression-free survival was 8.6 months with kelvaparib and 11.1 months with placebo (Table 11-5). The analysis included 203 of the 219 progression-free survival events the plan requires for the final analysis. In the planned primary analysis (stratified hazard ratio 1.77, 95 per cent CI 1.26 to 2.50; stratified log-rank p = 0.0009), progression-free survival was shorter with kelvaparib than with placebo, and the primary endpoint was not met. The unstratified analysis gave a hazard ratio of 1.62 (95 per cent CI 1.18 to 2.21; log-rank p = 0.0021). The overall response rate, a key secondary endpoint, was 157 of 336 subjects (46.7 per cent, 95 per cent CI (41.3, 52.2)). Because the primary endpoint was not met, the prespecified testing hierarchy stops there, and every secondary result in this report is descriptive.

Table 11-4 Best Overall Response by RECIST 1.1 – Full Analysis Set

Best overall response	Kelvaparib 300 mg (N=227)	Placebo (N=109)	Total (N=336)
Complete response (CR)	28 (12.3)	2 (1.8)	30 (8.9)
Partial response (PR)	101 (44.5)	26 (23.9)	127 (37.8)
Stable disease	57 (25.1)	52 (47.7)	109 (32.4)
Progressive disease	41 (18.1)	29 (26.6)	70 (20.8)
Not evaluable	0 (0.0)	0 (0.0)	0 (0.0)
Objective response rate, n (%)	129 (56.8)	28 (25.7)	157 (46.7)
95% CI	(50.1, 63.4)	(17.8, 34.9)	(41.3, 52.2)

CI = confidence interval. Assessed according to RECIST version 1.1. Confidence intervals are exact (Clopper-Pearson).

Primary endpoint. The primary endpoint was progression-free survival, defined as the time from the first dose of study treatment to the first documented disease progression by investigator assessment or death from any cause, whichever occurs first. The overall response rate, a key secondary endpoint, was 157 of 336 subjects (46.7 per cent, 95 per cent CI (41.3, 52.2)). Because the testing hierarchy is prespecified, significance for a secondary endpoint is claimed only where every preceding test reached significance; p-values below that point are reported as nominal. The estimate and its confidence interval are in Table 11-4. The interval rather than the point estimate is what should be read, because it is the interval that states how precisely this

study has answered the question.

Response. 157 of 336 subjects (46.7 per cent, 95 per cent confidence interval (41.3, 52.2)) had a response. Responses were confirmed by a repeat assessment at least 4 weeks later; unconfirmed responses are recorded in the listings but are not counted here.

Duration of response. The median was 11.4 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Progression-free survival. The median was 8.8 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Overall survival. The median was 28.7 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Consistency. The subgroup tables give response and median progression-free survival for all subjects in each subgroup, both treatment groups together; they do not compare the treatment groups within a subgroup, so they do not show whether the effect was consistent across subgroups. None of those analyses was powered and none is adjusted for multiplicity.

11.4.2 Analysis of time-to-event endpoints

Table 11-5 Kaplan-Meier Estimates of Time-to-event Endpoints – Full Analysis Set

Endpoint	Kelvaparib 300 mg (N=227)	Placebo (N=109)	Total (N=336)
Duration of response, median (months)	11.4	9.3	11.4
25th, 75th percentile	(6.1, 15.5)	(4.6, 19.6)	(5.4, 15.5)
Progression-free survival, median (months)	8.6	11.1	8.8
25th, 75th percentile	(2.8, 12.5)	(5.9, 21.2)	(3.7, 14.2)
Overall survival, median (months)	26.0	NE	28.7
25th, 75th percentile	(20.2, NE)	(21.9, NE)	(21.3, NE)

NE = not estimable. Duration of response is analysed among responding subjects only. It is measured from the first dose to the end of response recorded in the analysis dataset; response and progression dates cannot be re-derived because no assessment-level tumour data were collected in the database.

11.4.3 Subgroup analyses

Response and median progression-free survival are shown below for all subjects in each prespecified subgroup, both treatment groups together. These analyses were not powered, and no conclusion is drawn from any single subgroup.

Table 11-6 Overall Response Rate and Progression-free Survival by BRCA Mutation Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
BRCA wild-type	200	98 (49.0)	(41.9, 56.1)	9.1
BRCA1 mutated	81	38 (46.9)	(35.7, 58.3)	8.3
BRCA2 mutated	55	21 (38.2)	(25.4, 52.3)	9.2

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-7 Overall Response Rate and Progression-free Survival by Homologous Recombination Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Negative	164	75 (45.7)	(37.9, 53.7)	9.2
Positive	172	82 (47.7)	(40.0, 55.4)	8.8

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-8 Overall Response Rate and Progression-free Survival by Platinum Sensitivity – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
No	109	52 (47.7)	(38.1, 57.5)	8.6
Yes	227	105 (46.3)	(39.6, 53.0)	9.2

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-9 Overall Response Rate and Progression-free Survival by Residual Disease After Surgery – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Absent	196	94 (48.0)	(40.8, 55.2)	8.7
Present	140	63 (45.0)	(36.6, 53.6)	9.2

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-10 Overall Response Rate and Progression-free Survival by Prior Lines of Platinum-based Therapy – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
1 to 2	245	109 (44.5)	(38.2, 51.0)	8.7
>= 3	91	48 (52.7)	(42.0, 63.3)	8.8

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-11 Overall Response Rate and Progression-free Survival by Age Category – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
< 65 years	154	74 (48.1)	(39.9, 56.2)	9.1
>= 65 years	182	83 (45.6)	(38.2, 53.1)	8.8

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-12 Overall Response Rate and Progression-free Survival by ECOG Performance Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
0	142	62 (43.7)	(35.4, 52.2)	9.1
1	163	81 (49.7)	(41.8, 57.6)	8.6
2	31	14 (45.2)	(27.3, 64.0)	8.6

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

BRCA Mutation Status. BRCA wild-type: 98 of 200 (49.0 per cent); BRCA1 mutated: 38 of 81 (46.9 per cent); BRCA2 mutated: 21 of 55 (38.2 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Homologous Recombination Status. Negative: 75 of 164 (45.7 per cent); Positive: 82 of 172 (47.7 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Platinum Sensitivity. No: 52 of 109 (47.7 per cent); Yes: 105 of 227 (46.3 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Residual Disease After Surgery. Absent: 94 of 196 (48.0 per cent); Present: 63 of 140 (45.0 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Prior Lines of Platinum-based Therapy. 1 to 2: 109 of 245 (44.5 per cent); >= 3: 48 of 91 (52.7 per cent). The subgroups are of unequal size and none of these comparisons was powered, so

the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Age Category. < 65 years: 74 of 154 (48.1 per cent); >= 65 years: 83 of 182 (45.6 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

ECOG Performance Status. 0: 62 of 142 (43.7 per cent); 1: 81 of 163 (49.7 per cent); 2: 14 of 31 (45.2 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

11.4.4 Efficacy conclusions

Kelvaparib did not prolong progression-free survival compared with placebo in recurrent ovarian carcinoma, platinum-sensitive or platinum-resistant: progression-free survival was shorter with kelvaparib (stratified hazard ratio 1.77, 95 per cent CI 1.26 to 2.50; stratified log-rank p = 0.0009), and the primary endpoint was not met. The overall response rate of 46.7 per cent and every other secondary result are descriptive, because the testing hierarchy stops at the primary endpoint.

11.5 By-subject efficacy data

Individual response data for every subject are listed at Appendix 16.2.7, and individual time-to-event data at Appendix 16.2.8.

12 SAFETY EVALUATION

12.1 Extent of exposure

All 354 subjects received at least one dose. The extent of exposure is in Table 11-3.

12.2 Adverse events

12.2.1 Overview

Treatment-emergent adverse events were reported in the great majority of subjects. 179 subjects (50.6 per cent) had an event of grade 3 or higher, 83 (23.4 per cent) had a serious event and 43 (12.1 per cent) discontinued study treatment because of an event.

Table 12-1 Overview of Treatment-emergent Adverse Events – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Any TEAE	236 (100.0)	118 (100.0)	354 (100.0)
Any treatment-related TEAE	234 (99.2)	118 (100.0)	352 (99.4)
Any grade >= 3 TEAE	122 (51.7)	57 (48.3)	179 (50.6)
Any serious TEAE	57 (24.2)	26 (22.0)	83 (23.4)
Any TEAE leading to dose reduction	42 (17.8)	22 (18.6)	64 (18.1)
Any TEAE leading to discontinuation	30 (12.7)	13 (11.0)	43 (12.1)
Any fatal TEAE	22 (9.3)	6 (5.1)	28 (7.9)

TEAE = treatment-emergent adverse event. A subject is counted once within each category.

12.2.1.1 Discussion of each event term

Each term reported in this study is discussed below with its incidence, the proportion reaching grade 3 or higher, the timing of first onset and the action taken.

Nausea was reported in 223 subjects (63.0 per cent), of whom 5 (1.4 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 36 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 26 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Anaemia was reported in 191 subjects (54.0 per cent), of whom 62 (17.5 per cent of the study population) had an event of grade 3 or higher, and 36 had an event recorded as serious. The median time to first onset was 32 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 7 subjects. 3 events had a fatal outcome and are described in the narratives at Section 14.3.3. 40 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Fatigue was reported in 156 subjects (44.1 per cent), of whom 6 (1.7 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 32 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 23 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Vomiting was reported in 114 subjects (32.2 per cent), of whom 11 (3.1 per cent of the study population) had an event of grade 3 or higher, and 2 had an event recorded as serious. The

median time to first onset was 25 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 2 subjects. 14 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Constipation was reported in 112 subjects (31.6 per cent), of whom 2 (0.6 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 15 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Thrombocytopenia was reported in 95 subjects (26.8 per cent), of whom 11 (3.1 per cent of the study population) had an event of grade 3 or higher, and 2 had an event recorded as serious. The median time to first onset was 31 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 2 subjects. 2 events had a fatal outcome and are described in the narratives at Section 14.3.3. 14 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Decreased appetite was reported in 91 subjects (25.7 per cent), of whom 1 (0.3 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 31 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 20 subjects had an event that had not resolved at the data cut-off. The event falls within Metabolism and nutrition disorders.

Neutropenia was reported in 86 subjects (24.3 per cent), of whom 21 (5.9 per cent of the study population) had an event of grade 3 or higher, and 5 had an event recorded as serious. The median time to first onset was 40 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 2 subjects. 2 events had a fatal outcome and are described in the narratives at Section 14.3.3. 17 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Headache was reported in 76 subjects (21.5 per cent), of whom 2 (0.6 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 38 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 10 subjects had an event that had not resolved at the

data cut-off. The event falls within Nervous system disorders.

Dysgeusia was reported in 70 subjects (19.8 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 37 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 8 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Diarrhoea was reported in 69 subjects (19.5 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 25 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 12 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Asthenia was reported in 65 subjects (18.4 per cent), of whom 4 (1.1 per cent of the study population) had an event of grade 3 or higher, and 2 had an event recorded as serious. The median time to first onset was 18 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 10 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Dizziness was reported in 63 subjects (17.8 per cent), of whom 3 (0.8 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 29 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 9 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Abdominal pain was reported in 51 subjects (14.4 per cent), of whom 2 (0.6 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 39 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 9 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Oedema peripheral was reported in 46 subjects (13.0 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 36 days from the first dose, so the event arose within the period of most intensive monitoring in the

protocol. 5 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Arthralgia was reported in 45 subjects (12.7 per cent), of whom 2 (0.6 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 27 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 8 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Nasopharyngitis was reported in 43 subjects (12.1 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 35 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 8 subjects had an event that had not resolved at the data cut-off. The event falls within Infections and infestations.

Back pain was reported in 42 subjects (11.9 per cent), of whom 2 (0.6 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 20 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 8 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Cough was reported in 40 subjects (11.3 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 43 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 4 subjects had an event that had not resolved at the data cut-off. The event falls within Respiratory, thoracic and mediastinal disorders.

Aspartate aminotransferase increased was reported in 40 subjects (11.3 per cent), of whom 7 (2.0 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 43 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 8 subjects had an event that had not resolved at the data cut-off. The event falls within Investigations.

Insomnia was reported in 39 subjects (11.0 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from

the first dose, so the event arose within the period of most intensive monitoring in the protocol. 4 subjects had an event that had not resolved at the data cut-off. The event falls within Psychiatric disorders.

Pyrexia was reported in 35 subjects (9.9 per cent), of whom 1 (0.3 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 5 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Dyspepsia was reported in 33 subjects (9.3 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 37 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 8 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Myalgia was reported in 33 subjects (9.3 per cent), of whom 1 (0.3 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 42 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 7 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Alanine aminotransferase increased was reported in 32 subjects (9.0 per cent), of whom 4 (1.1 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 35 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Investigations.

Leukopenia was reported in 31 subjects (8.8 per cent), of whom 8 (2.3 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 5 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Urinary tract infection was reported in 30 subjects (8.5 per cent), of whom 4 (1.1 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 32 days from the first dose, so the event arose within the period of most intensive monitoring in the

protocol. It led to permanent discontinuation of study treatment in 1 subject. 5 subjects had an event that had not resolved at the data cut-off. The event falls within Infections and infestations.

Weight decreased was reported in 26 subjects (7.3 per cent), of whom 1 (0.3 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 41 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Investigations.

12.2.2 Events by system organ class and preferred term

Table 12-2 Treatment-emergent Adverse Events Occurring in at Least 10% of Subjects by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Gastrointestinal disorders	201 (85.2)	107 (90.7)	308 (87.0)
Nausea	147 (62.3)	76 (64.4)	223 (63.0)
Vomiting	72 (30.5)	42 (35.6)	114 (32.2)
Constipation	66 (28.0)	46 (39.0)	112 (31.6)
Diarrhoea	49 (20.8)	20 (16.9)	69 (19.5)
Abdominal pain	29 (12.3)	22 (18.6)	51 (14.4)
Blood and lymphatic system disorders	183 (77.5)	95 (80.5)	278 (78.5)
Anaemia	126 (53.4)	65 (55.1)	191 (54.0)
Thrombocytopenia	61 (25.8)	34 (28.8)	95 (26.8)
Neutropenia	55 (23.3)	31 (26.3)	86 (24.3)
General disorders and administration site conditions	144 (61.0)	88 (74.6)	232 (65.5)
Fatigue	94 (39.8)	62 (52.5)	156 (44.1)
Asthenia	38 (16.1)	27 (22.9)	65 (18.4)
Oedema peripheral	24 (10.2)	22 (18.6)	46 (13.0)
Nervous system disorders	111 (47.0)	59 (50.0)	170 (48.0)
Headache	54 (22.9)	22 (18.6)	76 (21.5)
Dysgeusia	41 (17.4)	29 (24.6)	70 (19.8)
Dizziness	42 (17.8)	21 (17.8)	63 (17.8)
Musculoskeletal and connective tissue disorders	67 (28.4)	39 (33.1)	106 (29.9)
Arthralgia	29 (12.3)	16 (13.6)	45 (12.7)
Back pain	25 (10.6)	17 (14.4)	42 (11.9)
Investigations	67 (28.4)	37 (31.4)	104 (29.4)
Aspartate aminotransferase increased	24 (10.2)	16 (13.6)	40 (11.3)
Infections and infestations	63 (26.7)	33 (28.0)	96 (27.1)
Nasopharyngitis	28 (11.9)	15 (12.7)	43 (12.1)
Metabolism and nutrition disorders	58 (24.6)	33 (28.0)	91 (25.7)
Decreased appetite	58 (24.6)	33 (28.0)	91 (25.7)
Respiratory, thoracic and mediastinal disorders	58 (24.6)	22 (18.6)	80 (22.6)
Cough	28 (11.9)	12 (10.2)	40 (11.3)
Psychiatric disorders	24 (10.2)	15 (12.7)	39 (11.0)
Insomnia	24 (10.2)	15 (12.7)	39 (11.0)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

Gastrointestinal disorders. Events in this class were reported in 308 subjects (87.0 per cent), and in 23 (6.5 per cent) at grade 3 or higher. The terms reported most often were nausea, vomiting, constipation. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Blood and lymphatic system disorders. Events in this class were reported in 278 subjects (78.5 per cent), and in 99 (28.0 per cent) at grade 3 or higher. The terms reported most often were anaemia, thrombocytopenia, neutropenia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

General disorders and administration site conditions. Events in this class were reported in 232 subjects (65.5 per cent), and in 11 (3.1 per cent) at grade 3 or higher. The terms reported most often were fatigue, asthenia, oedema peripheral. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Nervous system disorders. Events in this class were reported in 170 subjects (48.0 per cent), and in 5 (1.4 per cent) at grade 3 or higher. The terms reported most often were headache, dysgeusia, dizziness. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Musculoskeletal and connective tissue disorders. Events in this class were reported in 106 subjects (29.9 per cent), and in 5 (1.4 per cent) at grade 3 or higher. The terms reported most often were arthralgia, back pain, myalgia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Investigations. Events in this class were reported in 104 subjects (29.4 per cent), and in 14 (4.0 per cent) at grade 3 or higher. The terms reported most often were aspartate aminotransferase increased, alanine aminotransferase increased, weight decreased. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Infections and infestations. Events in this class were reported in 96 subjects (27.1 per cent), and in 21 (5.9 per cent) at grade 3 or higher. The terms reported most often were nasopharyngitis, urinary tract infection, pneumonia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Metabolism and nutrition disorders. Events in this class were reported in 91 subjects (25.7 per cent), and in 1 (0.3 per cent) at grade 3 or higher. The terms reported most often were decreased appetite. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Respiratory, thoracic and mediastinal disorders. Events in this class were reported in 80 subjects (22.6 per cent), and in 11 (3.1 per cent) at grade 3 or higher. The terms reported most often were cough, dyspnoea, pneumonitis. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Psychiatric disorders. Events in this class were reported in 39 subjects (11.0 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were insomnia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Skin and subcutaneous tissue disorders. Events in this class were reported in 25 subjects (7.1 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were rash. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Vascular disorders. Events in this class were reported in 17 subjects (4.8 per cent), and in 1 (0.3 per cent) at grade 3 or higher. The terms reported most often were hypertension. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Neoplasms benign, malignant and unspecified. Events in this class were reported in 14 subjects (4.0 per cent), and in 13 (3.7 per cent) at grade 3 or higher. The terms reported most often were myelodysplastic syndrome, malignant neoplasm progression. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

12.2.3 Events of grade 3 or higher

Table 12-3 Treatment-emergent Adverse Events of Grade 3 or Higher by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Blood and lymphatic system disorders	68 (28.8)	31 (26.3)	99 (28.0)
Anaemia	46 (19.5)	16 (13.6)	62 (17.5)
Gastrointestinal disorders	17 (7.2)	6 (5.1)	23 (6.5)
Infections and infestations	16 (6.8)	5 (4.2)	21 (5.9)
Investigations	9 (3.8)	5 (4.2)	14 (4.0)
Neoplasms benign, malignant and unspecified	5 (2.1)	8 (6.8)	13 (3.7)
Respiratory, thoracic and mediastinal disorders	11 (4.7)	0 (0.0)	11 (3.1)
General disorders and administration site conditions	4 (1.7)	7 (5.9)	11 (3.1)
Musculoskeletal and connective tissue disorders	4 (1.7)	1 (0.8)	5 (1.4)
Nervous system disorders	5 (2.1)	0 (0.0)	5 (1.4)
Metabolism and nutrition disorders	0 (0.0)	1 (0.8)	1 (0.3)
Vascular disorders	0 (0.0)	1 (0.8)	1 (0.3)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

12.2.4 Events considered related to study treatment

Table 12-4 Treatment-related Treatment-emergent Adverse Events Occurring in at Least 5% of Subjects – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Nausea	85 (36.0)	47 (39.8)	132 (37.3)
Anaemia	79 (33.5)	37 (31.4)	116 (32.8)
Fatigue	64 (27.1)	32 (27.1)	96 (27.1)
Constipation	41 (17.4)	30 (25.4)	71 (20.1)
Vomiting	40 (16.9)	27 (22.9)	67 (18.9)
Thrombocytopenia	39 (16.5)	22 (18.6)	61 (17.2)
Neutropenia	35 (14.8)	23 (19.5)	58 (16.4)
Decreased appetite	32 (13.6)	19 (16.1)	51 (14.4)
Headache	32 (13.6)	14 (11.9)	46 (13.0)
Dysgeusia	23 (9.7)	18 (15.3)	41 (11.6)
Dizziness	22 (9.3)	15 (12.7)	37 (10.5)
Asthenia	19 (8.1)	16 (13.6)	35 (9.9)
Diarrhoea	20 (8.5)	13 (11.0)	33 (9.3)
Back pain	19 (8.1)	13 (11.0)	32 (9.0)
Arthralgia	18 (7.6)	13 (11.0)	31 (8.8)
Oedema peripheral	15 (6.4)	15 (12.7)	30 (8.5)
Abdominal pain	15 (6.4)	15 (12.7)	30 (8.5)
Cough	18 (7.6)	8 (6.8)	26 (7.3)
Nasopharyngitis	18 (7.6)	7 (5.9)	25 (7.1)
Aspartate aminotransferase increased	13 (5.5)	12 (10.2)	25 (7.1)
Insomnia	13 (5.5)	11 (9.3)	24 (6.8)
Pyrexia	16 (6.8)	6 (5.1)	22 (6.2)
Alanine aminotransferase increased	16 (6.8)	6 (5.1)	22 (6.2)
Dyspepsia	11 (4.7)	8 (6.8)	19 (5.4)
Leukopenia	10 (4.2)	8 (6.8)	18 (5.1)

Relationship assessed by the investigator. Percentages are based on the number of subjects treated in each group. Overall N = 354.

12.2.5 Timing of onset

Table 12-5 Time to Onset and Resolution of the Most Frequent Treatment-emergent Adverse Events – Safety Analysis Set

Preferred term	Events, n	Median time to onset (days)	Resolved, n (%)	Median time to resolution (days)
Nausea	223	36.0	197 (88.3)	9.0
Anaemia	191	32.0	151 (79.1)	12.0
Fatigue	156	32.0	133 (85.3)	12.0
Vomiting	114	24.5	100 (87.7)	10.0
Constipation	112	32.5	97 (86.6)	9.0
Thrombocytopenia	95	31.0	81 (85.3)	10.0
Decreased appetite	91	31.0	71 (78.0)	9.0
Neutropenia	88	39.5	70 (79.5)	10.0
Headache	76	37.5	66 (86.8)	8.5
Dysgeusia	70	35.5	62 (88.6)	12.0
Diarrhoea	69	25.0	57 (82.6)	9.0
Asthenia	65	18.0	55 (84.6)	9.0
Dizziness	63	29.0	54 (85.7)	10.5
Abdominal pain	51	39.0	42 (82.4)	9.0
Oedema peripheral	46	35.5	41 (89.1)	10.0
Arthralgia	45	27.0	37 (82.2)	12.0
Nasopharyngitis	43	35.0	35 (81.4)	12.0
Back pain	42	18.0	34 (81.0)	11.5
Cough	40	42.5	36 (90.0)	12.0
Aspartate aminotransferase increased	40	41.5	32 (80.0)	12.5

Events are counted at the event level rather than the subject level. NE = not estimable.

12.2.6 Adverse events of special interest

Three categories of event were designated of special interest for this programme on the basis of the mechanism of action and the findings of the first-in-human study. Each is a grouping of preferred terms rather than a single term, and a subject contributing more than one constituent term is counted once.

Table 12-6 Adverse Events of Special Interest - Safety Analysis Set

Event of special interest	Minimum grade	Subjects, n (%)	Grade >= 3, n (%)	Serious, n (%)
Myelodysplastic syndrome or acute myeloid leukaemia	Any	8 (2.3)	7 (2.0)	6 (1.7)
Pneumonitis	Any	16 (4.5)	6 (1.7)	5 (1.4)
Severe anaemia	3	62 (17.5)	62 (17.5)	36 (10.2)

Constituent preferred terms. Myelodysplastic syndrome or acute myeloid leukaemia: Myelodysplastic syndrome; Pneumonitis: Pneumonitis; Severe anaemia: Anaemia. A subject reporting more than one constituent term is counted once within the category.

Myelodysplastic syndrome or acute myeloid leukaemia

Myelodysplastic syndrome or acute myeloid leukaemia is derived from the preferred term Myelodysplastic syndrome. The category was reported in 8 of 354 subjects (2.3 per cent), of whom 7 (2.0 per cent) had an event of grade 3 or higher and 1 had a fatal outcome.

The median time to first onset was 51 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Pneumonitis

Pneumonitis is reported as the coded preferred term. The category was reported in 16 of 354 subjects (4.5 per cent), of whom 6 (1.7 per cent) had an event of grade 3 or higher and 1 had a fatal outcome.

The median time to first onset was 53 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Severe anaemia

Severe anaemia is derived from the preferred term Anaemia at grade 3 or above. The category was reported in 62 of 354 subjects (17.5 per cent), of whom 62 (17.5 per cent) had an event of grade 3 or higher and 3 had a fatal outcome.

The median time to first onset was 32 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Nausea was the most frequent event overall, reported in 223 subjects (63.0 per cent), of whom 5 (1.4 per cent) had an event of grade 3 or higher.

12.3 Deaths, other serious adverse events and other significant events

12.3.1 Deaths

72 subjects died. The causes recorded are in Table 12-7. Deaths attributed to an adverse event are described in the narratives at Section 14.3.3.

Table 12-7 Deaths – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Any death	51 (21.6)	21 (17.8)	72 (20.3)
Death within 30 days of last dose	20 (8.5)	8 (6.8)	28 (7.9)
Primary cause of death			
Disease progression	24 (10.2)	15 (12.7)	39 (11.0)
Adverse event	22 (9.3)	6 (5.1)	28 (7.9)
Other	5 (2.1)	0 (0.0)	5 (1.4)

Percentages are based on the number of subjects treated within each group.

72 of 354 subjects (20.3 per cent) died. The primary cause recorded for each is below. Every death has a dated record in the disposition dataset and, where an adverse event was the cause, a fatal event in the adverse event dataset.

Disease progression. Recorded as the primary cause for 39 subjects (54.2 per cent of those who died). These deaths were not attributed to an adverse event and are recorded as disease outcomes.

Adverse event. Recorded as the primary cause for 28 subjects (38.9 per cent of those who died). Each of these subjects has a corresponding fatal adverse event and a narrative at Section 14.3.3.

Other. Recorded as the primary cause for 5 subjects (6.9 per cent of those who died). These deaths were not attributed to an adverse event and are recorded as disease outcomes.

12.3.2 Serious adverse events

Table 12-8 Serious Treatment-emergent Adverse Events by Preferred Term – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Anaemia	27 (11.4)	9 (7.6)	36 (10.2)
Malignant neoplasm progression	3 (1.3)	3 (2.5)	6 (1.7)
Myelodysplastic syndrome	2 (0.8)	4 (3.4)	6 (1.7)
Pneumonia	2 (0.8)	4 (3.4)	6 (1.7)
Neutropenia	3 (1.3)	2 (1.7)	5 (1.4)
Pneumonitis	5 (2.1)	0 (0.0)	5 (1.4)
Sepsis	4 (1.7)	0 (0.0)	4 (1.1)
Asthenia	1 (0.4)	1 (0.8)	2 (0.6)
Intestinal obstruction	2 (0.8)	0 (0.0)	2 (0.6)
Thrombocytopenia	2 (0.8)	0 (0.0)	2 (0.6)
Vomiting	2 (0.8)	0 (0.0)	2 (0.6)
Abdominal pain	0 (0.0)	1 (0.8)	1 (0.3)
Alanine aminotransferase increased	1 (0.4)	0 (0.0)	1 (0.3)
Arthralgia	1 (0.4)	0 (0.0)	1 (0.3)
Ascites	1 (0.4)	0 (0.0)	1 (0.3)
Aspartate aminotransferase increased	0 (0.0)	1 (0.8)	1 (0.3)
Blood creatinine increased	0 (0.0)	1 (0.8)	1 (0.3)
Dyspnoea	1 (0.4)	0 (0.0)	1 (0.3)
Headache	1 (0.4)	0 (0.0)	1 (0.3)
Pleural effusion	1 (0.4)	0 (0.0)	1 (0.3)
Pulmonary embolism	1 (0.4)	0 (0.0)	1 (0.3)

Subjects are counted once per preferred term. Percentages are based on N = 354.

12.3.3 Events leading to discontinuation

Table 12-9 Treatment-emergent Adverse Events Leading to Discontinuation of Study Treatment – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Anaemia	5 (2.1)	2 (1.7)	7 (2.0)
Malignant neoplasm progression	3 (1.3)	3 (2.5)	6 (1.7)
Myelodysplastic syndrome	1 (0.4)	3 (2.5)	4 (1.1)
Pneumonia	2 (0.8)	2 (1.7)	4 (1.1)
Sepsis	4 (1.7)	0 (0.0)	4 (1.1)
Intestinal obstruction	2 (0.8)	0 (0.0)	2 (0.6)
Neutropenia	2 (0.8)	0 (0.0)	2 (0.6)
Thrombocytopenia	2 (0.8)	0 (0.0)	2 (0.6)
Vomiting	1 (0.4)	1 (0.8)	2 (0.6)
Abdominal pain	1 (0.4)	0 (0.0)	1 (0.3)
Ascites	1 (0.4)	0 (0.0)	1 (0.3)
Aspartate aminotransferase increased	1 (0.4)	0 (0.0)	1 (0.3)
Asthenia	0 (0.0)	1 (0.8)	1 (0.3)
Back pain	0 (0.0)	1 (0.8)	1 (0.3)
Dyspnoea	1 (0.4)	0 (0.0)	1 (0.3)
Pleural effusion	1 (0.4)	0 (0.0)	1 (0.3)
Pneumonitis	1 (0.4)	0 (0.0)	1 (0.3)
Pulmonary embolism	1 (0.4)	0 (0.0)	1 (0.3)
Urinary tract infection	1 (0.4)	0 (0.0)	1 (0.3)

Percentages are based on N = 354.

12.3.4 Narratives

Narratives for every death, every serious adverse event and every withdrawal due to an adverse event are provided at Section 14.3.3, bound separately.

12.4 Clinical laboratory evaluation

Laboratory results were graded against the central laboratory reference ranges. Shifts from baseline to the worst post-baseline grade are in Table 12-10. Individual laboratory data are listed at Appendix 16.2.10.

Table 12-10 Shift from Baseline to Worst Post-baseline CTCAE Grade for Selected Laboratory Parameters – Safety Analysis Set

Laboratory parameter	Kelvaparib 300 mg: worst grade 3 n (%)	Kelvaparib 300 mg: worst grade 4 n (%)	Kelvaparib 300 mg: shift >= 2 grades n (%)	Placebo: worst grade 3 n (%)	Placebo: worst grade 4 n (%)	Placebo: shift >= 2 grades n (%)	Total: worst grade 3 n (%)	Total: worst grade 4 n (%)	Total: shift >= 2 grades n (%)
Haemoglobin	46 (19.5)	5 (2.1)	89 (37.7)	13 (11.0)	5 (4.2)	33 (28.0)	59 (16.7)	10 (2.8)	122 (34.5)
Platelets	26 (11.0)	7 (3.0)	51 (21.6)	10 (8.5)	1 (0.8)	17 (14.4)	36 (10.2)	8 (2.3)	68 (19.2)
Neutrophils	24 (10.2)	6 (2.5)	54 (22.9)	13 (11.0)	1 (0.8)	25 (21.2)	37 (10.5)	7 (2.0)	79 (22.3)
Lymphocytes	19 (8.1)	2 (0.8)	40 (16.9)	7 (5.9)	1 (0.8)	22 (18.6)	26 (7.3)	3 (0.8)	62 (17.5)
CA-125	34 (14.4)	15 (6.4)	74 (31.4)	23 (19.5)	1 (0.8)	45 (38.1)	57 (16.1)	16 (4.5)	119 (33.6)
ALT	11 (4.7)	5 (2.1)	26 (11.0)	6 (5.1)	1 (0.8)	16 (13.6)	17 (4.8)	6 (1.7)	42 (11.9)
AST	11 (4.7)	3 (1.3)	18 (7.6)	3 (2.5)	1 (0.8)	6 (5.1)	14 (4.0)	4 (1.1)	24 (6.8)
Creatinine	10 (4.2)	3 (1.3)	31 (13.1)	4 (3.4)	3 (2.5)	14 (11.9)	14 (4.0)	6 (1.7)	45 (12.7)
Lactate dehydrogenase	8 (3.4)	5 (2.1)	30 (12.7)	3 (2.5)	3 (2.5)	15 (12.7)	11 (3.1)	8 (2.3)	45 (12.7)
Albumin	6 (2.5)	5 (2.1)	23 (9.7)	4 (3.4)	1 (0.8)	13 (11.0)	10 (2.8)	6 (1.7)	36 (10.2)

CTCAE = Common Terminology Criteria for Adverse Events, version 5.0. Percentages are based on the number of subjects treated in each group. Overall N = 354.

12.4.1 Findings for each laboratory test

ALT. An abnormality of any grade was present at baseline in 93 subjects (26.3 per cent) and post-baseline in 149 (42.1 per cent). 80 subjects (22.6 per cent) shifted to a worse grade than their own baseline, and 23 (6.5 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

AST. An abnormality of any grade was present at baseline in 110 subjects (31.1 per cent) and post-baseline in 151 (42.7 per cent). 65 subjects (18.4 per cent) shifted to a worse grade than their own baseline, and 18 (5.1 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Albumin. An abnormality of any grade was present at baseline in 97 subjects (27.4 per cent) and post-baseline in 148 (41.8 per cent). 78 subjects (22.0 per cent) shifted to a worse grade than their own baseline, and 16 (4.5 per cent) reached grade 3 or higher at some point. Values were

reviewed against the dose modification rules at every visit at which they were measured.

CA-125. An abnormality of any grade was present at baseline in 109 subjects (30.8 per cent) and post-baseline in 279 (78.8 per cent). 249 subjects (70.3 per cent) shifted to a worse grade than their own baseline, and 73 (20.6 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Creatinine. An abnormality of any grade was present at baseline in 111 subjects (31.4 per cent) and post-baseline in 172 (48.6 per cent). 80 subjects (22.6 per cent) shifted to a worse grade than their own baseline, and 20 (5.6 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Haemoglobin. An abnormality of any grade was present at baseline in 92 subjects (26.0 per cent) and post-baseline in 283 (79.9 per cent). 257 subjects (72.6 per cent) shifted to a worse grade than their own baseline, and 69 (19.5 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Lactate dehydrogenase. An abnormality of any grade was present at baseline in 96 subjects (27.1 per cent) and post-baseline in 167 (47.2 per cent). 101 subjects (28.5 per cent) shifted to a worse grade than their own baseline, and 19 (5.4 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Lymphocytes. An abnormality of any grade was present at baseline in 97 subjects (27.4 per cent) and post-baseline in 199 (56.2 per cent). 127 subjects (35.9 per cent) shifted to a worse grade than their own baseline, and 29 (8.2 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Neutrophils. An abnormality of any grade was present at baseline in 92 subjects (26.0 per cent) and post-baseline in 215 (60.7 per cent). 170 subjects (48.0 per cent) shifted to a worse grade than their own baseline, and 44 (12.4 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Platelets. An abnormality of any grade was present at baseline in 122 subjects (34.5 per cent) and post-baseline in 230 (65.0 per cent). 169 subjects (47.7 per cent) shifted to a worse grade than their own baseline, and 44 (12.4 per cent) reached grade 3 or higher at some point. Values

were reviewed against the dose modification rules at every visit at which they were measured.

12.5 Vital signs, physical findings and other safety observations

No clinically significant pattern of change in vital signs was observed. Individual vital sign data are listed at Appendix 16.2.11.

12.5.1 Electrocardiograms

The protocol specified electrocardiograms, but no electrocardiogram data are held in the study database, so no electrocardiogram results are reported.

12.6 Safety conclusions

The safety profile observed was consistent with that anticipated for oral PARP1-selective inhibitor in this population. The events that occurred most often, and those that most often reached grade 3 or higher, are those for which the protocol required specific monitoring and management. No new safety concern was identified.

13 DISCUSSION AND OVERALL CONCLUSIONS

Study KLV-7703 evaluated kelvaparib in recurrent ovarian carcinoma, platinum-sensitive or platinum-resistant. Kelvaparib did not prolong progression-free survival compared with placebo in recurrent ovarian carcinoma, platinum-sensitive or platinum-resistant: progression-free survival was shorter with kelvaparib (stratified hazard ratio 1.77, 95 per cent CI 1.26 to 2.50; stratified log-rank p = 0.0009), and the primary endpoint was not met. The overall response rate of 46.7 per cent and every other secondary result are descriptive, because the testing hierarchy stops at the primary endpoint. The safety profile was dominated by anaemia and myelosuppression, both on-target consequences of the mechanism of action. Myelodysplastic syndrome or acute myeloid leukaemia occurred in 2.3 per cent of subjects (Section 12.2.6). Limitations should be borne in mind. The study was open-label, which may influence investigator assessment of response and reporting of subjective adverse events. Follow-up was ongoing at the data cut-off and time-to-event estimates are therefore immature. Taken together, study KLV-7703 did not show a progression-free survival benefit for kelvaparib at 300 mg in recurrent ovarian carcinoma, platinum-sensitive or platinum-resistant.

The population enrolled was consistent with the intended population and with the eligibility criteria. Exposure was sufficient to characterise both the efficacy and the safety of the regimen. The primary analysis was performed as planned; the sensitivity analyses the statistical analysis plan lists are not reported here.

The population enrolled was the population the protocol described. Median age, performance status and prior therapy were consistent with the eligibility criteria, and there is no indication that the study enrolled a population easier to treat than the one intended.

Exposure was adequate. The median relative dose intensity indicates that the regimen as written could be delivered, and the pattern of reduction and interruption was consistent with the toxicities anticipated for oral PARP1-selective inhibitor.

The efficacy result should be read together with its confidence interval and with the maturity of the follow-up. Because the study was randomised, the comparison is internally controlled and the estimate does not depend on an external assumption about how this population would otherwise have fared.

The safety profile was consistent with the mechanism and with the earlier clinical experience. The events reported most often, and those most often reaching grade 3 or higher, were those for

which the protocol required specific monitoring, which indicates that the monitoring was directed at the right things.

43 subjects (12.1 per cent) discontinued study treatment because of an adverse event. That figure, rather than the overall incidence of events, is the measure of whether the regimen is tolerable in practice, because it counts the events that could not be managed by dose modification.

Deaths were reviewed individually. Those attributed to an adverse event are described in the narratives; the remainder are recorded as disease outcomes and are consistent with what would be expected in this population over this period.

The limitations of this study should be borne in mind. Follow-up at this analysis is limited and the longer time-to-event estimates are immature. The study was open-label, so investigator assessment of any subjective endpoint may have been influenced by knowledge of the assigned treatment. The subgroup analyses are descriptive.

Taken together, the results support the conclusions stated above and do not identify a safety concern that was not anticipated. The design assumed a hazard ratio of 0.63; the observed hazard ratio was above 1, so the assumption about the treatment effect was not borne out.

The limitations of the study should be borne in mind when interpreting these results. The study was open-label, so investigator assessment of subjective endpoints may have been influenced by knowledge of the assigned treatment. Follow-up at the time of this analysis is limited, so the estimates of the longer time-to-event endpoints are immature and will change as further events accrue.

14 TABLES AND FIGURES REFERRED TO BUT NOT IN THE TEXT

The summary tables referred to in the text are presented in this section. Subject data listings are provided in Appendix 16.2.

14.1 Demographic data

Table 14.1.1 Demographic and Baseline Characteristics – Safety Analysis Set

Characteristic	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Age (years)			
Mean (SD)	65.0 (8.44)	66.4 (8.93)	65.5 (8.62)
Median	65.0	66.5	65.0
Min, max	44, 88	45, 88	44, 88
>= 65 years, n (%)	125 (53.0)	69 (58.5)	194 (54.8)
Sex, n (%)			
Male	0 (0.0)	0 (0.0)	0 (0.0)
Female	236 (100.0)	118 (100.0)	354 (100.0)
Race, n (%)			
White	192 (81.4)	98 (83.1)	290 (81.9)
Asian	19 (8.1)	6 (5.1)	25 (7.1)
Black or African American	16 (6.8)	6 (5.1)	22 (6.2)
Other	9 (3.8)	8 (6.8)	17 (4.8)
ECOG performance status, n (%)			
0	98 (41.5)	50 (42.4)	148 (41.8)
1	118 (50.0)	53 (44.9)	171 (48.3)
2	20 (8.5)	15 (12.7)	35 (9.9)
BRCA mutation status, n (%)			
BRCA1 mutated	55 (23.3)	29 (24.6)	84 (23.7)
BRCA2 mutated	44 (18.6)	14 (11.9)	58 (16.4)
BRCA wild-type	137 (58.1)	75 (63.6)	212 (59.9)
Homologous recombination deficiency, n (%)	122 (51.7)	63 (53.4)	185 (52.3)
Platinum-sensitive disease, n (%)	155 (65.7)	82 (69.5)	237 (66.9)
Residual disease after cytoreductive surgery, n (%)	101 (42.8)	46 (39.0)	147 (41.5)
Number of prior lines of platinum-based therapy			
Median	2.0	2.0	2.0
>= 3, n (%)	64 (27.1)	33 (28.0)	97 (27.4)

SD = standard deviation; ECOG = Eastern Cooperative Oncology Group. Percentages are based on the number of subjects in the safety analysis set within each group.

Table 14.1.2 Subject Disposition – Safety Analysis Set

Disposition	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Subjects treated, n	236	118	354
Treatment ongoing at data cut-off, n (%)	61 (25.8)	21 (17.8)	82 (23.2)
Discontinued study treatment, n (%)	175 (74.2)	97 (82.2)	272 (76.8)
Disease progression	40 (16.9)	26 (22.0)	66 (18.6)
Adverse event	11 (4.7)	6 (5.1)	17 (4.8)
Death	20 (8.5)	8 (6.8)	28 (7.9)
Withdrawal by subject	37 (15.7)	24 (20.3)	61 (17.2)
Physician decision	35 (14.8)	19 (16.1)	54 (15.3)
Lost to follow-up	17 (7.2)	6 (5.1)	23 (6.5)
Protocol deviation	15 (6.4)	8 (6.8)	23 (6.5)

Percentages are based on the number of subjects treated within each group.

Table 14.1.3 Extent of Exposure – Safety Analysis Set

Exposure parameter	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Duration of treatment (days)			
Median	134.0	127.5	131.5
Q1, Q3	82.0, 181.8	73.8, 165.5	80.0, 179.0
Min, max	8, 307	3, 280	3, 307
Cycles started, median	5.0	4.5	5.0
Relative dose intensity (%), median	85.7	85.9	85.8
Any dose reduction, n (%)	42 (17.8)	23 (19.5)	65 (18.4)
Any dose interruption, n (%)	51 (21.6)	32 (27.1)	83 (23.4)

Q1 = first quartile; Q3 = third quartile. Relative dose intensity is the total dose received divided by the total dose planned over the treatment period.

14.2 Efficacy data

Table 14.2.1 Best Overall Response by RECIST 1.1 – Full Analysis Set

Best overall response	Kelvaparib 300 mg (N=227)	Placebo (N=109)	Total (N=336)
Complete response (CR)	28 (12.3)	2 (1.8)	30 (8.9)
Partial response (PR)	101 (44.5)	26 (23.9)	127 (37.8)
Stable disease	57 (25.1)	52 (47.7)	109 (32.4)
Progressive disease	41 (18.1)	29 (26.6)	70 (20.8)
Not evaluable	0 (0.0)	0 (0.0)	0 (0.0)
Objective response rate, n (%)	129 (56.8)	28 (25.7)	157 (46.7)
95% CI	(50.1, 63.4)	(17.8, 34.9)	(41.3, 52.2)

CI = confidence interval. Assessed according to RECIST version 1.1. Confidence intervals are exact (Clopper-Pearson).

Table 14.2.2 Kaplan-Meier Estimates of Time-to-event Endpoints – Full Analysis Set

Endpoint	Kelvaparib 300 mg (N=227)	Placebo (N=109)	Total (N=336)
Duration of response, median (months)	11.4	9.3	11.4
25th, 75th percentile	(6.1, 15.5)	(4.6, 19.6)	(5.4, 15.5)
Progression-free survival, median (months)	8.6	11.1	8.8
25th, 75th percentile	(2.8, 12.5)	(5.9, 21.2)	(3.7, 14.2)
Overall survival, median (months)	26.0	NE	28.7
25th, 75th percentile	(20.2, NE)	(21.9, NE)	(21.3, NE)

NE = not estimable. Duration of response is analysed among responding subjects only. It is measured from the first dose to the end of response recorded in the analysis dataset; response and progression dates cannot be re-derived because no assessment-level tumour data were collected in the database.

Table 14.2.3 Overall Response Rate and Progression-free Survival by BRCA Mutation Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
BRCA wild-type	200	98 (49.0)	(41.9, 56.1)	9.1
BRCA1 mutated	81	38 (46.9)	(35.7, 58.3)	8.3
BRCA2 mutated	55	21 (38.2)	(25.4, 52.3)	9.2

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.4 Overall Response Rate and Progression-free Survival by Homologous Recombination Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Negative	164	75 (45.7)	(37.9, 53.7)	9.2
Positive	172	82 (47.7)	(40.0, 55.4)	8.8

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.5 Overall Response Rate and Progression-free Survival by Platinum Sensitivity – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
No	109	52 (47.7)	(38.1, 57.5)	8.6
Yes	227	105 (46.3)	(39.6, 53.0)	9.2

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.6 Overall Response Rate and Progression-free Survival by Residual Disease After Surgery – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Absent	196	94 (48.0)	(40.8, 55.2)	8.7
Present	140	63 (45.0)	(36.6, 53.6)	9.2

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.7 Overall Response Rate and Progression-free Survival by Prior Lines of Platinum-based Therapy – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
1 to 2	245	109 (44.5)	(38.2, 51.0)	8.7
>= 3	91	48 (52.7)	(42.0, 63.3)	8.8

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.8 Overall Response Rate and Progression-free Survival by Age Category – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
< 65 years	154	74 (48.1)	(39.9, 56.2)	9.1
>= 65 years	182	83 (45.6)	(38.2, 53.1)	8.8

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.9 Overall Response Rate and Progression-free Survival by ECOG Performance Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
0	142	62 (43.7)	(35.4, 52.2)	9.1
1	163	81 (49.7)	(41.8, 57.6)	8.6
2	31	14 (45.2)	(27.3, 64.0)	8.6

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

14.3 Safety data

Table 14.3.1 Overview of Treatment-emergent Adverse Events – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Any TEAE	236 (100.0)	118 (100.0)	354 (100.0)
Any treatment-related TEAE	234 (99.2)	118 (100.0)	352 (99.4)
Any grade >= 3 TEAE	122 (51.7)	57 (48.3)	179 (50.6)
Any serious TEAE	57 (24.2)	26 (22.0)	83 (23.4)
Any TEAE leading to dose reduction	42 (17.8)	22 (18.6)	64 (18.1)
Any TEAE leading to discontinuation	30 (12.7)	13 (11.0)	43 (12.1)
Any fatal TEAE	22 (9.3)	6 (5.1)	28 (7.9)

TEAE = treatment-emergent adverse event. A subject is counted once within each category.

Table 14.3.2 Treatment-emergent Adverse Events Occurring in at Least 2% of Subjects by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Gastrointestinal disorders	201 (85.2)	107 (90.7)	308 (87.0)
Nausea	147 (62.3)	76 (64.4)	223 (63.0)
Vomiting	72 (30.5)	42 (35.6)	114 (32.2)
Constipation	66 (28.0)	46 (39.0)	112 (31.6)
Diarrhoea	49 (20.8)	20 (16.9)	69 (19.5)
Abdominal pain	29 (12.3)	22 (18.6)	51 (14.4)
Dyspepsia	21 (8.9)	12 (10.2)	33 (9.3)
Blood and lymphatic system disorders	183 (77.5)	95 (80.5)	278 (78.5)
Anaemia	126 (53.4)	65 (55.1)	191 (54.0)
Thrombocytopenia	61 (25.8)	34 (28.8)	95 (26.8)
Neutropenia	55 (23.3)	31 (26.3)	86 (24.3)
Leukopenia	16 (6.8)	15 (12.7)	31 (8.8)
Lymphopenia	20 (8.5)	4 (3.4)	24 (6.8)
General disorders and administration site conditions	144 (61.0)	88 (74.6)	232 (65.5)
Fatigue	94 (39.8)	62 (52.5)	156 (44.1)
Asthenia	38 (16.1)	27 (22.9)	65 (18.4)
Oedema peripheral	24 (10.2)	22 (18.6)	46 (13.0)
Pyrexia	25 (10.6)	10 (8.5)	35 (9.9)
Nervous system disorders	111 (47.0)	59 (50.0)	170 (48.0)
Headache	54 (22.9)	22 (18.6)	76 (21.5)
Dysgeusia	41 (17.4)	29 (24.6)	70 (19.8)
Dizziness	42 (17.8)	21 (17.8)	63 (17.8)
Musculoskeletal and connective tissue disorders	67 (28.4)	39 (33.1)	106 (29.9)
Arthralgia	29 (12.3)	16 (13.6)	45 (12.7)
Back pain	25 (10.6)	17 (14.4)	42 (11.9)
Myalgia	17 (7.2)	16 (13.6)	33 (9.3)
Investigations	67 (28.4)	37 (31.4)	104 (29.4)
Aspartate aminotransferase increased	24 (10.2)	16 (13.6)	40 (11.3)
Alanine aminotransferase increased	23 (9.7)	9 (7.6)	32 (9.0)
Weight decreased	19 (8.1)	7 (5.9)	26 (7.3)
Blood creatinine increased	15 (6.4)	10 (8.5)	25 (7.1)
Infections and infestations	63 (26.7)	33 (28.0)	96 (27.1)
Nasopharyngitis	28 (11.9)	15 (12.7)	43 (12.1)
Urinary tract infection	19 (8.1)	11 (9.3)	30 (8.5)

System organ class / Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Pneumonia	14 (5.9)	7 (5.9)	21 (5.9)
Sepsis	8 (3.4)	1 (0.8)	9 (2.5)
Metabolism and nutrition disorders	58 (24.6)	33 (28.0)	91 (25.7)
Decreased appetite	58 (24.6)	33 (28.0)	91 (25.7)
Respiratory, thoracic and mediastinal disorders	58 (24.6)	22 (18.6)	80 (22.6)
Cough	28 (11.9)	12 (10.2)	40 (11.3)
Dyspnoea	18 (7.6)	6 (5.1)	24 (6.8)
Pneumonitis	12 (5.1)	4 (3.4)	16 (4.5)
Psychiatric disorders	24 (10.2)	15 (12.7)	39 (11.0)
Insomnia	24 (10.2)	15 (12.7)	39 (11.0)
Skin and subcutaneous tissue disorders	16 (6.8)	9 (7.6)	25 (7.1)
Rash	16 (6.8)	9 (7.6)	25 (7.1)
Vascular disorders	13 (5.5)	4 (3.4)	17 (4.8)
Hypertension	13 (5.5)	4 (3.4)	17 (4.8)
Neoplasms benign, malignant and unspecified	6 (2.5)	8 (6.8)	14 (4.0)
Myelodysplastic syndrome	3 (1.3)	5 (4.2)	8 (2.3)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

Table 14.3.3 Treatment-emergent Adverse Events of Grade 3 or Higher by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Blood and lymphatic system disorders	68 (28.8)	31 (26.3)	99 (28.0)
Anaemia	46 (19.5)	16 (13.6)	62 (17.5)
Neutropenia	12 (5.1)	9 (7.6)	21 (5.9)
Thrombocytopenia	7 (3.0)	4 (3.4)	11 (3.1)
Leukopenia	5 (2.1)	3 (2.5)	8 (2.3)
Lymphopenia	3 (1.3)	2 (1.7)	5 (1.4)
Gastrointestinal disorders	17 (7.2)	6 (5.1)	23 (6.5)
Vomiting	8 (3.4)	3 (2.5)	11 (3.1)
Nausea	4 (1.7)	1 (0.8)	5 (1.4)
Intestinal obstruction	2 (0.8)	0 (0.0)	2 (0.6)
Abdominal pain	1 (0.4)	1 (0.8)	2 (0.6)
Constipation	1 (0.4)	1 (0.8)	2 (0.6)
Ascites	1 (0.4)	0 (0.0)	1 (0.3)
Infections and infestations	16 (6.8)	5 (4.2)	21 (5.9)
Pneumonia	6 (2.5)	5 (4.2)	11 (3.1)
Sepsis	6 (2.5)	0 (0.0)	6 (1.7)
Urinary tract infection	4 (1.7)	0 (0.0)	4 (1.1)
Investigations	9 (3.8)	5 (4.2)	14 (4.0)
Aspartate aminotransferase increased	3 (1.3)	4 (3.4)	7 (2.0)
Alanine aminotransferase increased	4 (1.7)	0 (0.0)	4 (1.1)
Blood creatinine increased	1 (0.4)	1 (0.8)	2 (0.6)
Weight decreased	1 (0.4)	0 (0.0)	1 (0.3)
Neoplasms benign, malignant and unspecified	5 (2.1)	8 (6.8)	13 (3.7)
Myelodysplastic syndrome	2 (0.8)	5 (4.2)	7 (2.0)
Malignant neoplasm progression	3 (1.3)	3 (2.5)	6 (1.7)
Respiratory, thoracic and mediastinal disorders	11 (4.7)	0 (0.0)	11 (3.1)
Pneumonitis	6 (2.5)	0 (0.0)	6 (1.7)
Dyspnoea	3 (1.3)	0 (0.0)	3 (0.8)
Pleural effusion	1 (0.4)	0 (0.0)	1 (0.3)
Pulmonary embolism	1 (0.4)	0 (0.0)	1 (0.3)
General disorders and administration site conditions	4 (1.7)	7 (5.9)	11 (3.1)
Fatigue	2 (0.8)	4 (3.4)	6 (1.7)
Asthenia	2 (0.8)	2 (1.7)	4 (1.1)

System organ class / Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Pyrexia	0 (0.0)	1 (0.8)	1 (0.3)
Musculoskeletal and connective tissue disorders	4 (1.7)	1 (0.8)	5 (1.4)
Arthralgia	2 (0.8)	0 (0.0)	2 (0.6)
Back pain	1 (0.4)	1 (0.8)	2 (0.6)
Myalgia	1 (0.4)	0 (0.0)	1 (0.3)
Nervous system disorders	5 (2.1)	0 (0.0)	5 (1.4)
Dizziness	3 (1.3)	0 (0.0)	3 (0.8)
Headache	2 (0.8)	0 (0.0)	2 (0.6)
Metabolism and nutrition disorders	0 (0.0)	1 (0.8)	1 (0.3)
Decreased appetite	0 (0.0)	1 (0.8)	1 (0.3)
Vascular disorders	0 (0.0)	1 (0.8)	1 (0.3)
Hypertension	0 (0.0)	1 (0.8)	1 (0.3)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

Table 14.3.4 Treatment-related Treatment-emergent Adverse Events Occurring in at Least 2% of Subjects – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Nausea	85 (36.0)	47 (39.8)	132 (37.3)
Anaemia	79 (33.5)	37 (31.4)	116 (32.8)
Fatigue	64 (27.1)	32 (27.1)	96 (27.1)
Constipation	41 (17.4)	30 (25.4)	71 (20.1)
Vomiting	40 (16.9)	27 (22.9)	67 (18.9)
Thrombocytopenia	39 (16.5)	22 (18.6)	61 (17.2)
Neutropenia	35 (14.8)	23 (19.5)	58 (16.4)
Decreased appetite	32 (13.6)	19 (16.1)	51 (14.4)
Headache	32 (13.6)	14 (11.9)	46 (13.0)
Dysgeusia	23 (9.7)	18 (15.3)	41 (11.6)
Dizziness	22 (9.3)	15 (12.7)	37 (10.5)
Asthenia	19 (8.1)	16 (13.6)	35 (9.9)
Diarrhoea	20 (8.5)	13 (11.0)	33 (9.3)
Back pain	19 (8.1)	13 (11.0)	32 (9.0)
Arthralgia	18 (7.6)	13 (11.0)	31 (8.8)
Oedema peripheral	15 (6.4)	15 (12.7)	30 (8.5)
Abdominal pain	15 (6.4)	15 (12.7)	30 (8.5)
Cough	18 (7.6)	8 (6.8)	26 (7.3)
Nasopharyngitis	18 (7.6)	7 (5.9)	25 (7.1)
Aspartate aminotransferase increased	13 (5.5)	12 (10.2)	25 (7.1)
Insomnia	13 (5.5)	11 (9.3)	24 (6.8)
Pyrexia	16 (6.8)	6 (5.1)	22 (6.2)
Alanine aminotransferase increased	16 (6.8)	6 (5.1)	22 (6.2)
Dyspepsia	11 (4.7)	8 (6.8)	19 (5.4)
Leukopenia	10 (4.2)	8 (6.8)	18 (5.1)
Dyspnoea	12 (5.1)	5 (4.2)	17 (4.8)
Urinary tract infection	10 (4.2)	7 (5.9)	17 (4.8)
Blood creatinine increased	10 (4.2)	7 (5.9)	17 (4.8)
Myalgia	7 (3.0)	10 (8.5)	17 (4.8)
Weight decreased	10 (4.2)	6 (5.1)	16 (4.5)
Rash	11 (4.7)	4 (3.4)	15 (4.2)
Hypertension	11 (4.7)	2 (1.7)	13 (3.7)
Pneumonia	9 (3.8)	3 (2.5)	12 (3.4)
Lymphopenia	9 (3.8)	3 (2.5)	12 (3.4)
Pneumonitis	9 (3.8)	3 (2.5)	12 (3.4)

Relationship assessed by the investigator. Percentages are based on the number of subjects treated in each group. Overall N = 354.

Table 14.3.5 Serious Treatment-emergent Adverse Events by Preferred Term – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Anaemia	27 (11.4)	9 (7.6)	36 (10.2)
Malignant neoplasm progression	3 (1.3)	3 (2.5)	6 (1.7)
Myelodysplastic syndrome	2 (0.8)	4 (3.4)	6 (1.7)
Pneumonia	2 (0.8)	4 (3.4)	6 (1.7)
Neutropenia	3 (1.3)	2 (1.7)	5 (1.4)
Pneumonitis	5 (2.1)	0 (0.0)	5 (1.4)
Sepsis	4 (1.7)	0 (0.0)	4 (1.1)
Asthenia	1 (0.4)	1 (0.8)	2 (0.6)
Intestinal obstruction	2 (0.8)	0 (0.0)	2 (0.6)
Thrombocytopenia	2 (0.8)	0 (0.0)	2 (0.6)
Vomiting	2 (0.8)	0 (0.0)	2 (0.6)
Abdominal pain	0 (0.0)	1 (0.8)	1 (0.3)
Alanine aminotransferase increased	1 (0.4)	0 (0.0)	1 (0.3)
Arthralgia	1 (0.4)	0 (0.0)	1 (0.3)
Ascites	1 (0.4)	0 (0.0)	1 (0.3)
Aspartate aminotransferase increased	0 (0.0)	1 (0.8)	1 (0.3)
Blood creatinine increased	0 (0.0)	1 (0.8)	1 (0.3)
Dyspnoea	1 (0.4)	0 (0.0)	1 (0.3)
Headache	1 (0.4)	0 (0.0)	1 (0.3)
Pleural effusion	1 (0.4)	0 (0.0)	1 (0.3)
Pulmonary embolism	1 (0.4)	0 (0.0)	1 (0.3)

Subjects are counted once per preferred term. Percentages are based on N = 354.

Table 14.3.6 Treatment-emergent Adverse Events Leading to Discontinuation of Study Treatment – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Anaemia	5 (2.1)	2 (1.7)	7 (2.0)
Malignant neoplasm progression	3 (1.3)	3 (2.5)	6 (1.7)
Myelodysplastic syndrome	1 (0.4)	3 (2.5)	4 (1.1)
Pneumonia	2 (0.8)	2 (1.7)	4 (1.1)
Sepsis	4 (1.7)	0 (0.0)	4 (1.1)
Intestinal obstruction	2 (0.8)	0 (0.0)	2 (0.6)
Neutropenia	2 (0.8)	0 (0.0)	2 (0.6)
Thrombocytopenia	2 (0.8)	0 (0.0)	2 (0.6)
Vomiting	1 (0.4)	1 (0.8)	2 (0.6)
Abdominal pain	1 (0.4)	0 (0.0)	1 (0.3)
Ascites	1 (0.4)	0 (0.0)	1 (0.3)
Aspartate aminotransferase increased	1 (0.4)	0 (0.0)	1 (0.3)
Asthenia	0 (0.0)	1 (0.8)	1 (0.3)
Back pain	0 (0.0)	1 (0.8)	1 (0.3)
Dyspnoea	1 (0.4)	0 (0.0)	1 (0.3)
Pleural effusion	1 (0.4)	0 (0.0)	1 (0.3)
Pneumonitis	1 (0.4)	0 (0.0)	1 (0.3)
Pulmonary embolism	1 (0.4)	0 (0.0)	1 (0.3)
Urinary tract infection	1 (0.4)	0 (0.0)	1 (0.3)

Percentages are based on N = 354.

Table 14.3.7 Deaths – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=236)	Placebo (N=118)	Total (N=354)
Any death	51 (21.6)	21 (17.8)	72 (20.3)
Death within 30 days of last dose	20 (8.5)	8 (6.8)	28 (7.9)
Primary cause of death			
Disease progression	24 (10.2)	15 (12.7)	39 (11.0)
Adverse event	22 (9.3)	6 (5.1)	28 (7.9)
Other	5 (2.1)	0 (0.0)	5 (1.4)

Percentages are based on the number of subjects treated within each group.

Table 14.3.8 Shift from Baseline to Worst Post-baseline CTCAE Grade for Selected Laboratory Parameters – Safety Analysis Set

Laboratory parameter	Kelvaparib 300 mg: worst grade 3 n (%)	Kelvaparib 300 mg: worst grade 4 n (%)	Kelvaparib 300 mg: shift >= 2 grades n (%)	Placebo: worst grade 3 n (%)	Placebo: worst grade 4 n (%)	Placebo: shift >= 2 grades n (%)	Total: worst grade 3 n (%)	Total: worst grade 4 n (%)	Total: shift >= 2 grades n (%)
Haemoglobin	46 (19.5)	5 (2.1)	89 (37.7)	13 (11.0)	5 (4.2)	33 (28.0)	59 (16.7)	10 (2.8)	122 (34.5)
Platelets	26 (11.0)	7 (3.0)	51 (21.6)	10 (8.5)	1 (0.8)	17 (14.4)	36 (10.2)	8 (2.3)	68 (19.2)
Neutrophils	24 (10.2)	6 (2.5)	54 (22.9)	13 (11.0)	1 (0.8)	25 (21.2)	37 (10.5)	7 (2.0)	79 (22.3)
Lymphocytes	19 (8.1)	2 (0.8)	40 (16.9)	7 (5.9)	1 (0.8)	22 (18.6)	26 (7.3)	3 (0.8)	62 (17.5)
CA-125	34 (14.4)	15 (6.4)	74 (31.4)	23 (19.5)	1 (0.8)	45 (38.1)	57 (16.1)	16 (4.5)	119 (33.6)
ALT	11 (4.7)	5 (2.1)	26 (11.0)	6 (5.1)	1 (0.8)	16 (13.6)	17 (4.8)	6 (1.7)	42 (11.9)
AST	11 (4.7)	3 (1.3)	18 (7.6)	3 (2.5)	1 (0.8)	6 (5.1)	14 (4.0)	4 (1.1)	24 (6.8)
Creatinine	10 (4.2)	3 (1.3)	31 (13.1)	4 (3.4)	3 (2.5)	14 (11.9)	14 (4.0)	6 (1.7)	45 (12.7)
Lactate dehydrogenase	8 (3.4)	5 (2.1)	30 (12.7)	3 (2.5)	3 (2.5)	15 (12.7)	11 (3.1)	8 (2.3)	45 (12.7)
Albumin	6 (2.5)	5 (2.1)	23 (9.7)	4 (3.4)	1 (0.8)	13 (11.0)	10 (2.8)	6 (1.7)	36 (10.2)

CTCAE = Common Terminology Criteria for Adverse Events, version 5.0. Percentages are based on the number of subjects treated in each group. Overall N = 354.

Table 14.3.9 Time to Onset and Resolution of the Most Frequent Treatment-emergent Adverse Events – Safety Analysis Set

Preferred term	Events, n	Median time to onset (days)	Resolved, n (%)	Median time to resolution (days)
Nausea	223	36.0	197 (88.3)	9.0
Anaemia	191	32.0	151 (79.1)	12.0
Fatigue	156	32.0	133 (85.3)	12.0
Vomiting	114	24.5	100 (87.7)	10.0
Constipation	112	32.5	97 (86.6)	9.0
Thrombocytopenia	95	31.0	81 (85.3)	10.0
Decreased appetite	91	31.0	71 (78.0)	9.0
Neutropenia	88	39.5	70 (79.5)	10.0
Headache	76	37.5	66 (86.8)	8.5
Dysgeusia	70	35.5	62 (88.6)	12.0
Diarrhoea	69	25.0	57 (82.6)	9.0
Asthenia	65	18.0	55 (84.6)	9.0
Dizziness	63	29.0	54 (85.7)	10.5
Abdominal pain	51	39.0	42 (82.4)	9.0
Oedema peripheral	46	35.5	41 (89.1)	10.0
Arthralgia	45	27.0	37 (82.2)	12.0
Nasopharyngitis	43	35.0	35 (81.4)	12.0
Back pain	42	18.0	34 (81.0)	11.5
Cough	40	42.5	36 (90.0)	12.0
Aspartate aminotransferase increased	40	41.5	32 (80.0)	12.5

Events are counted at the event level rather than the subject level. NE = not estimable.

14.3.10 Adverse events within subgroups

The summaries in Tables 14.3.10 to 14.3.16 are exploratory. The subgroups were not chosen or sized for comparisons of safety, the tables are not adjusted for multiplicity, and no conclusion about a difference between subgroups is drawn from them.

Table 14.3.10 Overview of adverse events by BRCA mutation status

Category	BRCA wild-type	BRCA1 mutated	BRCA2 mutated
Any event	212 (100.0)	84 (100.0)	58 (100.0)
Grade 3 or higher	110 (51.9)	40 (47.6)	29 (50.0)
Serious	46 (21.7)	23 (27.4)	14 (24.1)
Led to discontinuation	21 (9.9)	13 (15.5)	9 (15.5)
Fatal	10 (4.7)	10 (11.9)	8 (13.8)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.11 Overview of adverse events by homologous recombination status

Category	Negative	Positive
Any event	169 (100.0)	185 (100.0)
Grade 3 or higher	73 (43.2)	106 (57.3)
Serious	30 (17.8)	53 (28.6)
Led to discontinuation	11 (6.5)	32 (17.3)
Fatal	9 (5.3)	19 (10.3)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.12 Overview of adverse events by platinum sensitivity

Category	No	Yes
Any event	117 (100.0)	237 (100.0)
Grade 3 or higher	67 (57.3)	112 (47.3)
Serious	31 (26.5)	52 (21.9)
Led to discontinuation	17 (14.5)	26 (11.0)
Fatal	12 (10.3)	16 (6.8)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.13 Overview of adverse events by residual disease after surgery

Category	Absent	Present
Any event	207 (100.0)	147 (100.0)
Grade 3 or higher	106 (51.2)	73 (49.7)
Serious	49 (23.7)	34 (23.1)
Led to discontinuation	29 (14.0)	14 (9.5)
Fatal	15 (7.2)	13 (8.8)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.14 Overview of adverse events by prior lines of platinum-based therapy

Category	1 to 2	>= 3
Any event	257 (100.0)	97 (100.0)
Grade 3 or higher	137 (53.3)	42 (43.3)
Serious	65 (25.3)	18 (18.6)
Led to discontinuation	35 (13.6)	8 (8.2)
Fatal	25 (9.7)	3 (3.1)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.15 Overview of adverse events by age category

Category	<65	>=65
Any event	160 (100.0)	194 (100.0)
Grade 3 or higher	81 (50.6)	98 (50.5)
Serious	34 (21.2)	49 (25.3)
Led to discontinuation	18 (11.2)	25 (12.9)
Fatal	14 (8.8)	14 (7.2)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.16 Overview of adverse events by ECOG performance status

Category	0	1	2
Any event	148 (100.0)	171 (100.0)	35 (100.0)
Grade 3 or higher	69 (46.6)	93 (54.4)	17 (48.6)
Serious	32 (21.6)	38 (22.2)	13 (37.1)
Led to discontinuation	15 (10.1)	22 (12.9)	6 (17.1)
Fatal	12 (8.1)	14 (8.2)	2 (5.7)

Percentages are calculated on the number of subjects in that subgroup.

14.4 Pharmacokinetic data

Table 14.4.1 Summary of Steady-state Pharmacokinetic Parameters – Pharmacokinetic Analysis Set

Treatment	n	Cmax (ng/mL) Geometric mean (%CV)	tmax (h) Median (min, max)	AUC0-12 (ng·h/mL) Geometric mean (%CV)	t1/2 (h) Mean (SD)	CL/F (L/h) Geometric mean (%CV)
Kelvaparib 300 mg	236	1765 (28.4)	2.0 (2.0, 2.0)	17719 (35.0)	14.5 (6.24)	16.93 (35.0)

AUC0-12 = area under the concentration-time curve over one dosing interval; CL/F = apparent oral clearance; Cmax = maximum observed concentration; %CV = geometric coefficient of variation; t1/2 = terminal elimination half-life; tmax = time to maximum concentration.

Table 14.4.2 Accumulation Ratio and Trough Concentration – Pharmacokinetic Analysis Set

Treatment	n	Accumulation ratio Geometric mean (%CV)	Ctrough (ng/mL) Geometric mean (%CV)
Kelvaparib 300 mg	236	2.86 (24.1)	1093.9 (49.0)

Accumulation ratio is the ratio of AUC0-24 at steady state to AUC0-24 after the first dose.

15 REFERENCE LIST

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16 APPENDICES

16.1.1	Protocol and protocol amendments
16.1.2	Sample case report form
16.1.3	List of independent ethics committees and institutional review boards
16.1.4	List of investigators and study sites
16.1.5	Signatures of principal investigators
16.1.6	Listing of subjects receiving test drug from specific batches
16.1.7	Randomisation scheme and codes
16.1.8	Audit certificates
16.1.9	Documentation of statistical methods
16.1.10	Documentation of inter-laboratory standardisation and quality assurance
16.1.11	Publications based on the study
16.1.12	Important publications referenced in the report
16.2.1	Listing of subjects who discontinued study treatment
16.2.2	Listing of protocol deviations
16.2.3	Listing of subjects excluded from the efficacy analysis
16.2.4	Listing of demographic data
16.2.5	Listing of study treatment administration and compliance
16.2.6	Listing of individual drug concentration data
16.2.7	Listing of individual efficacy response data
16.2.8	Listing of individual time to event data
16.2.9	Listing of adverse events
16.2.10	Listing of individual laboratory measurements
16.2.11	Listing of vital signs
16.2.12	Listing of concomitant medications
16.2.13	Listing of subject disposition
16.3	Case report forms (2 volumes)
16.4	Individual subject data listings (archival): not applicable; individual data are provided as the analysis datasets